Local Decoder.

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Introduction

Today:

- Proof.
    1. Local Decoders.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M : (0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority are local decoders.

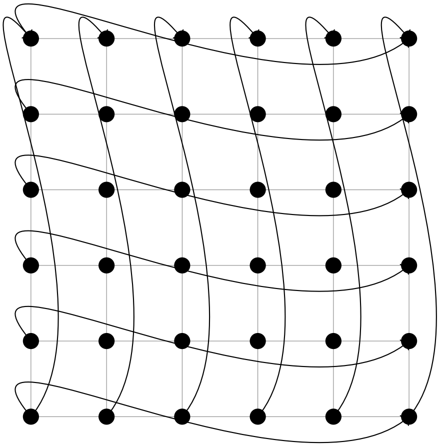
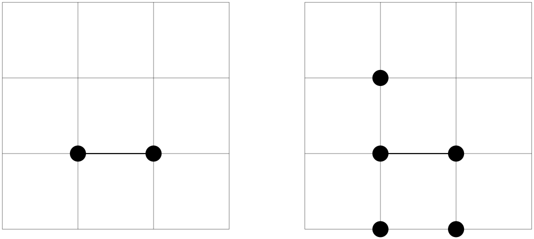


Figure: On the left is the Toric Graph. On the right are cross and face checks.



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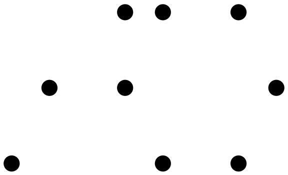

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